

💡 mod 0 'da handshaking yok (Basic I/O)

mod 1 'de handshaking tabanlı veri letişimi.

Input için strobe bekler, veri bulunduğu anda IBF set olur.

Output için OBF yollar, Ack bekler.

mod 2 'de handshaking var, çift yönlü haberleşme.

BSR 'de ilgili bitler üzerinde I/O ve bitlerin strobe, ack, OBF, IBF için kullanılır. Bu set / reset işlemleri PCi için yapılabilir.

MP
with

Status Word

PC'den yapılan okumalara karşılık gelmektedir. OBF, IBF, INTR ile I/O için kullanılan PCi'lere erişilebilir.

= 8255 mod 1 (Status Word)

• Input

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
I/O	I/O	IBF _A	INTE _A	INTR _A	INTE _B	IBF _B	INTR _B
Group A				Group B			

• output

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
OBF _A	INTE _A	I/O	I/O	INTR _A	INTE _B	OBF _B	INTR _B
Group A				Group B			

Mod 2

• Strobed bidirectional bus I/O

• Sadece grup A mod 2 destekler

• 1 Çift yönlü Port (PA) ve 5 handshaking kontrol işareti.

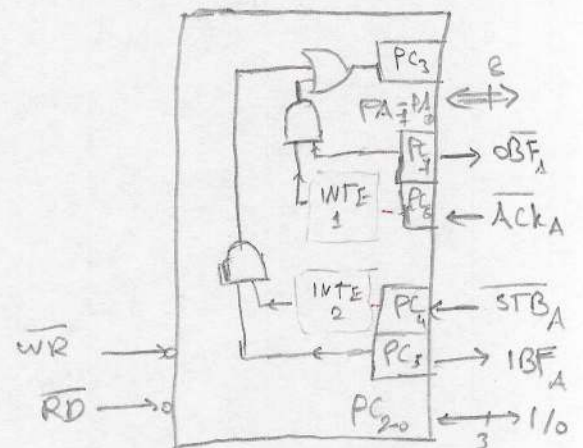
Ex CW

D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀
1	1	X	X	X	I/O	I/O	I/O

↑ BSR mod not set
↑ mod 2
↑ No dir. in mod 2

→ PC₂₋₄
1: input
0: output

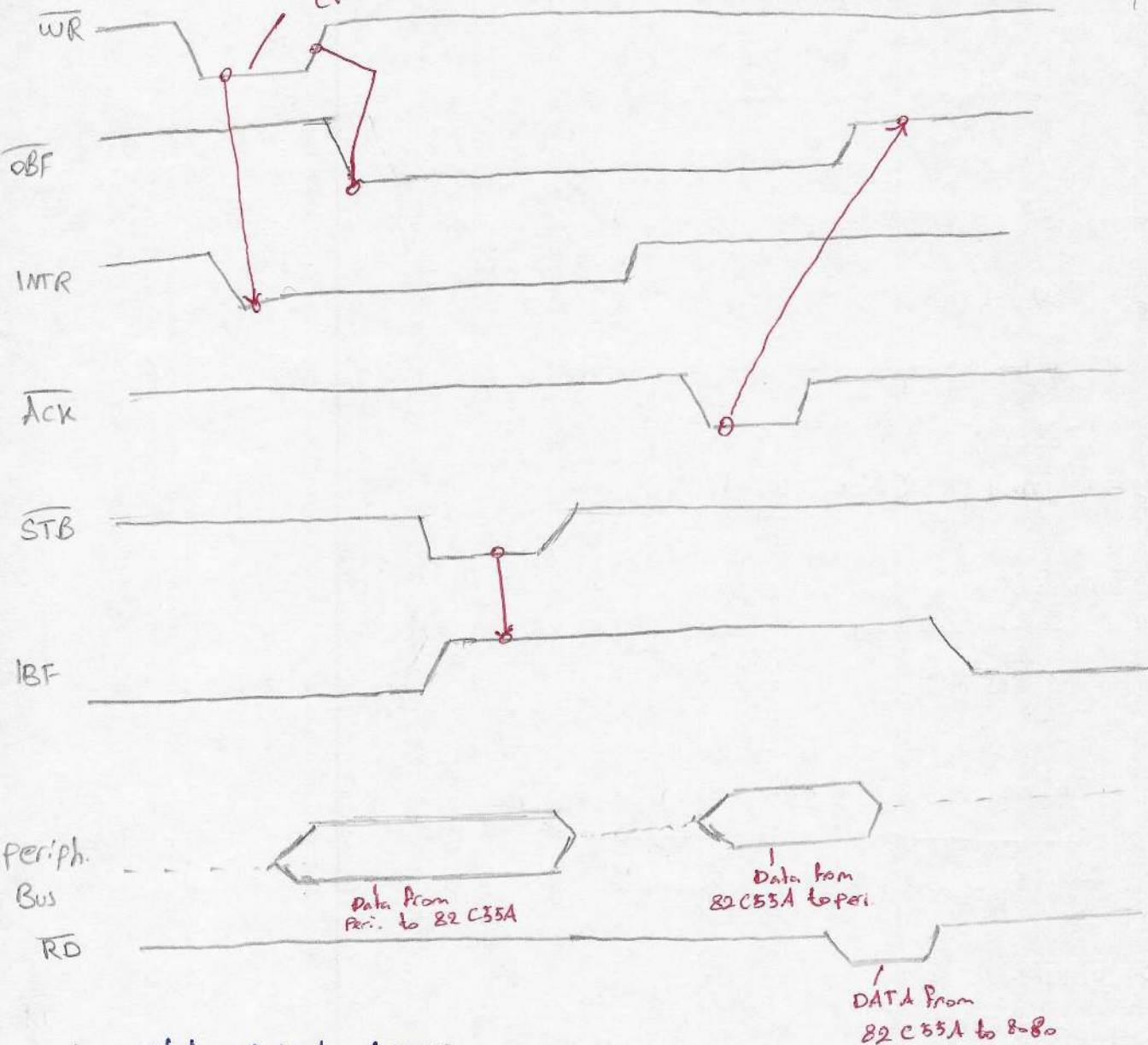
→ PB
1: input
0: output



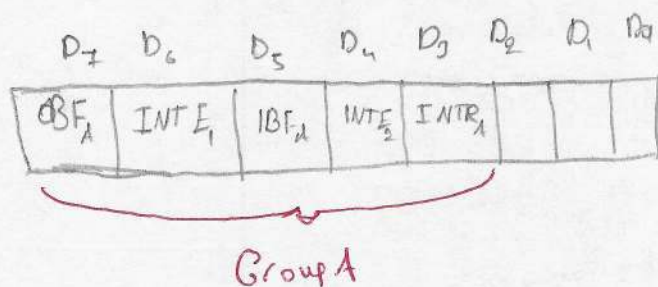
... BSR modunda INT set edilir

8255 Mod2 - Timing

HP
W3/2



mod2 - Status Word (8255)



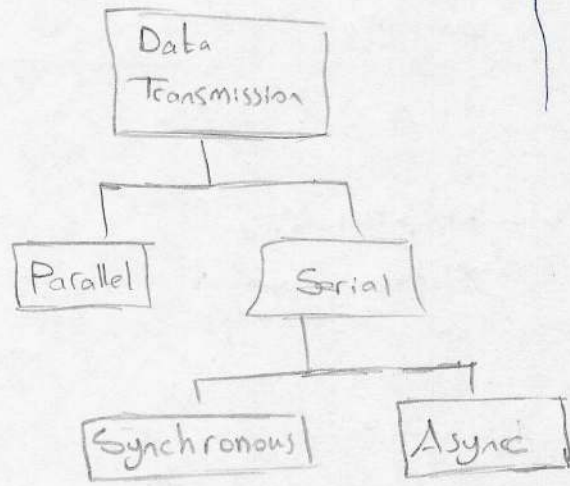
Versi iletisi:

• Seri:

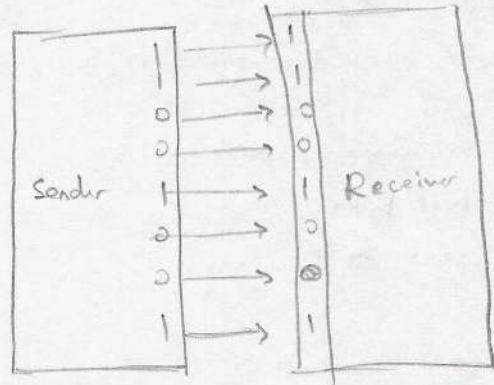
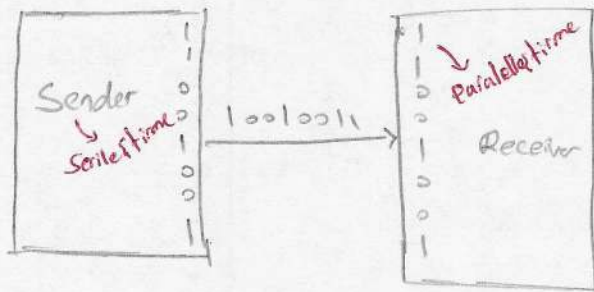
- Daha az maliyet
- Daha yavaş

• Paralel

- Daha hızlı
- Kısa mesafeler için



mp
w7/13

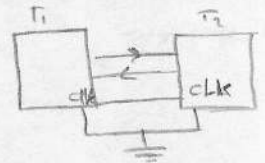


• Senkron

+ ortak bir saat darbesi (Done in hardware using phase locked loops (PLLs))

+ Block of data can be sent

+ More efficient: Less overhead than async transmission



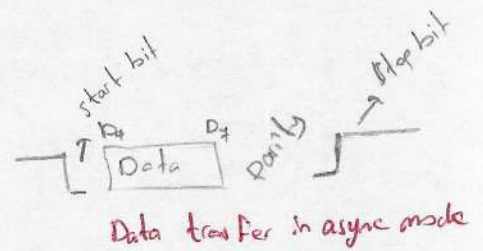
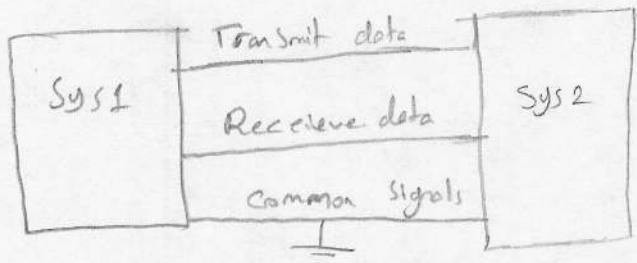
• Asenkron (baudrate, data length, parity are chosen beforehand)

+ Each byte encoded for transmission (start and stop bits)

+ Bir frekansta anlaşıp iletişim başlıyor

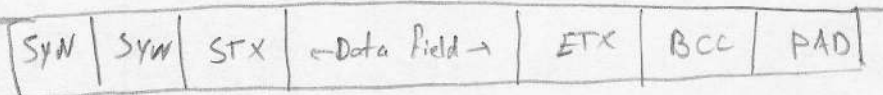
Farklılıklar olsa da kısıt paketlerdeki başlangıç ve bitiş bitlerinden dolayı telafi ediliyor.

+ ortak CLK yok



💡 each bit is accepted if sampled 2 times.
Recall Nyquist!

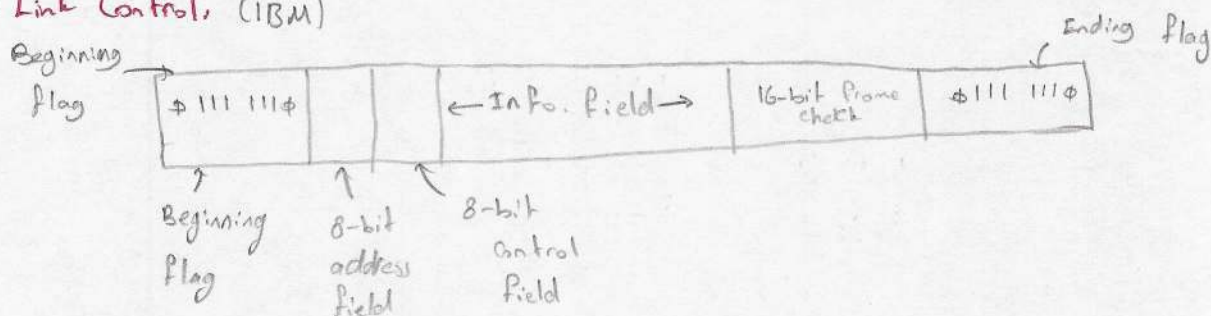
Bi-sync:



size of block can be ≥ 100 . BCC checks for errors

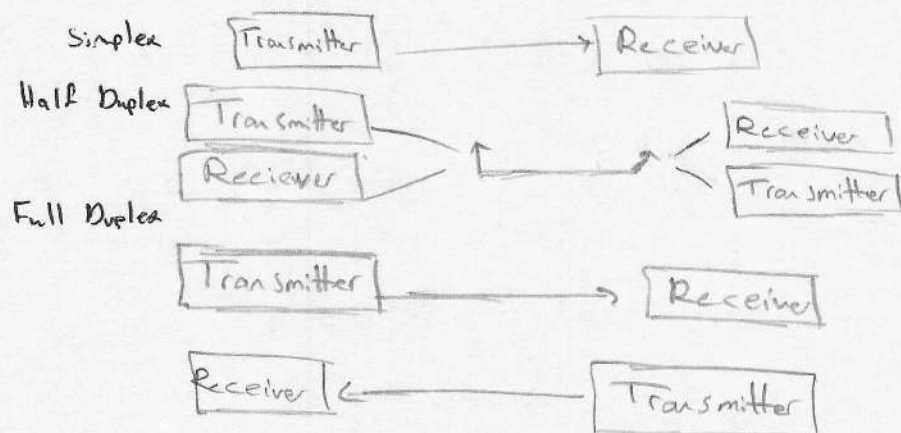
MP
w 7/4

Serial Data Link Control, (IBM)



Simplex / Duplex:

- Simplex: Data transmitted in one direction
- Duplex: Data flow in both directions
 - half duplex: Only one direction at a time
 - Full duplex: Two directions at the same time



Transmission Rate

Rate at which bits are transmitted (BAUD).

= Number of signal changes per second

e.g. 1200 baud means $\frac{1}{1200}$ seconds

Time the bit must be set

Ex what is the data rate in bit/sec and char rate if bit time is 3.33 ms (1 start, 8 data, 2 stop)

- Bit rate = $1 / 3.33 \text{ ms} = 300 \text{ bit/sec}$

- $11 \times 3.33 \text{ ms} = 36.63 \text{ ms}$ required to transmit a char so char rate = $1 / 36.63 \text{ ms} = 27.3 \text{ char/sec}$

Error Check

MP
W3/5

Parity check

- Even parity: When odd number of 1s set $D7=1$

Check Sum

Sum of all bytes without carry and 2's complements

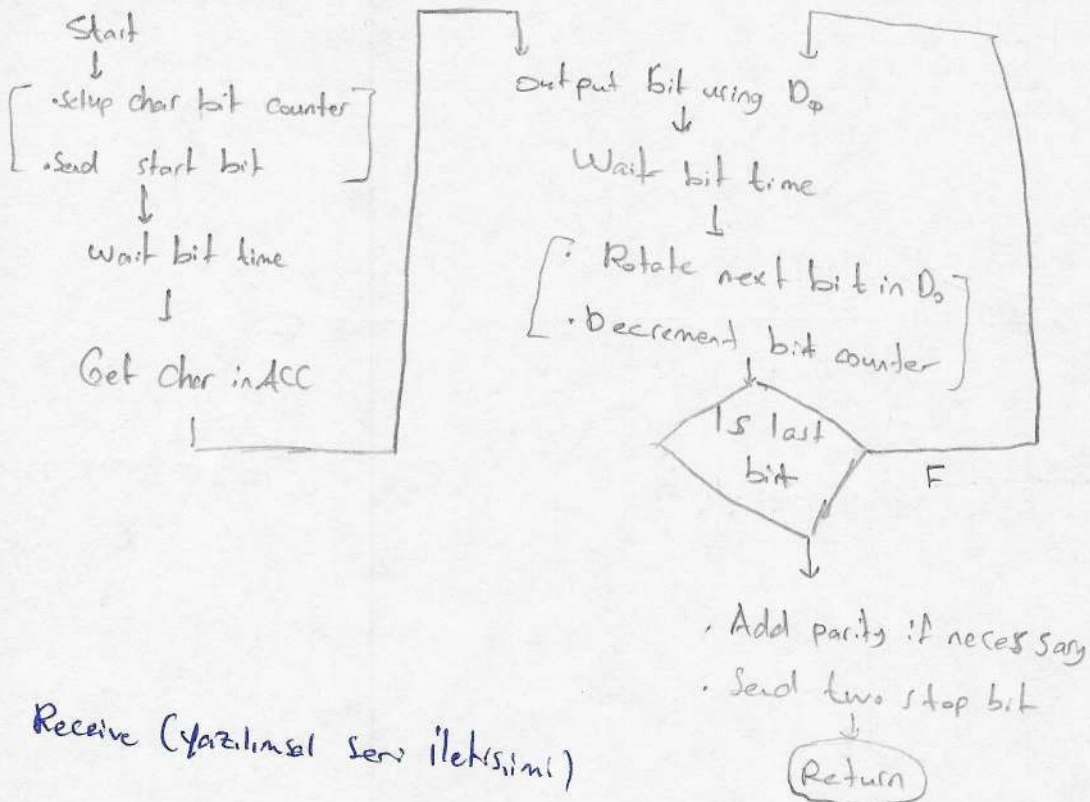
- Total sum should be zero

Cyclic Redundancy Code (CRC)

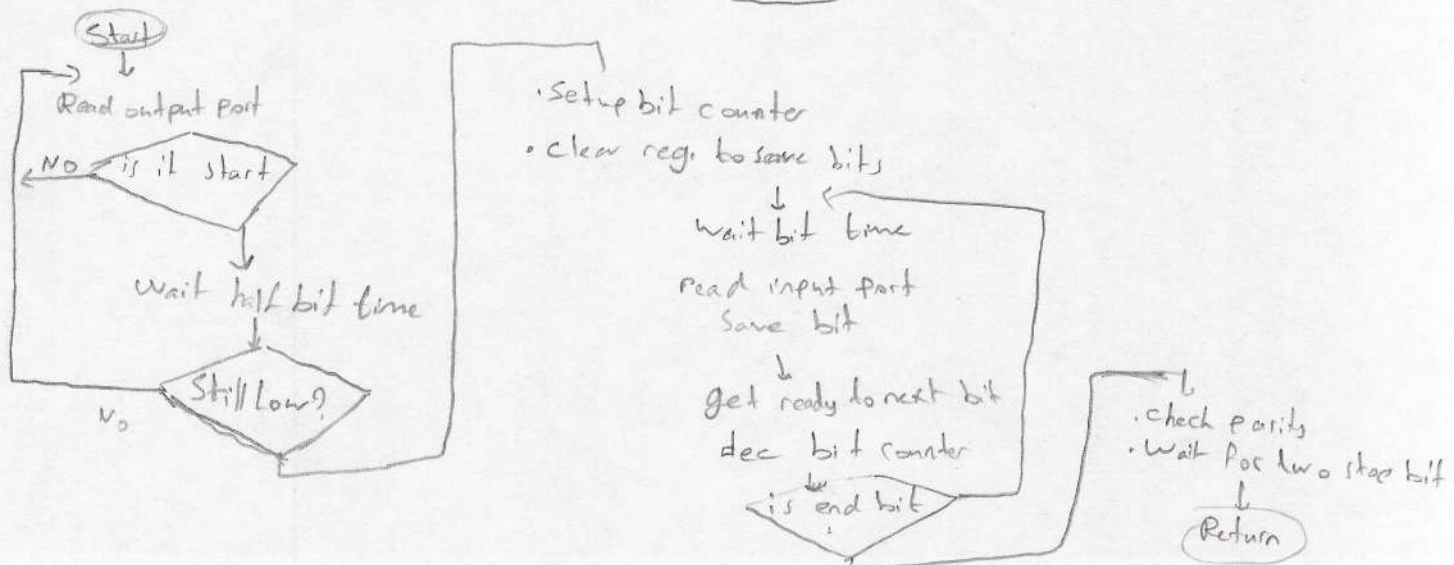
- Sync. Comm.

- Stream of data represented by cyclic polynomial divided by a constant polynomial.
- Remainder to set Bits and send out to check for error.

Transmit (Yazılımsal Seri iletişimi)



Receive (Yazılımsal Seri iletişimi)



8251 USART (universal sync/asynchronous receiver/transmitter)

Sends data received bit by bit.

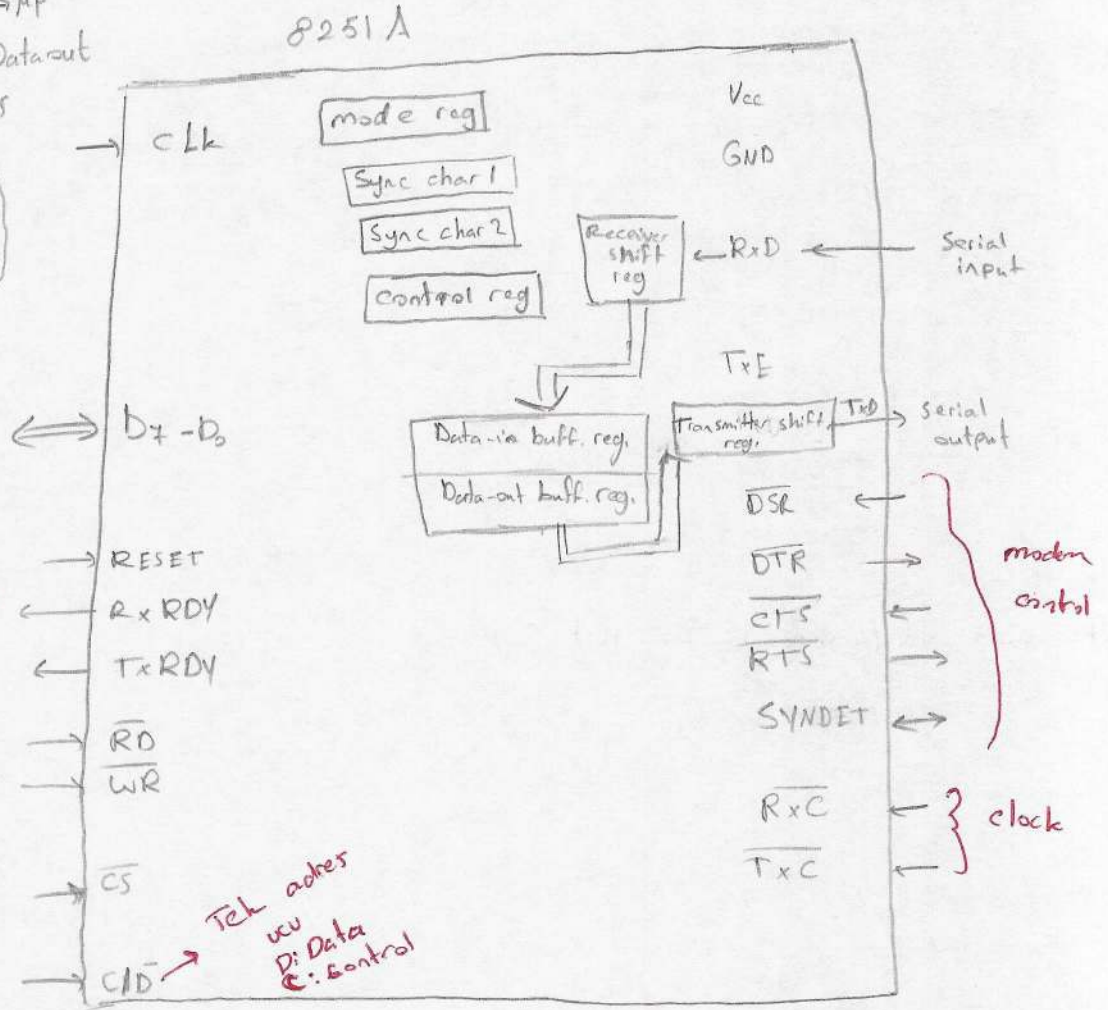
MP
w716

$C/\overline{D}=0, \overline{RD}=0$ Datain $\rightarrow$ MP

$C/\overline{D}=0, \overline{WR}=0$ MP $\rightarrow$ Dataout

$C/\overline{D}=1, \overline{RD}=0$ Status

$C/\overline{D}=1, \overline{WR}=0$ mode
Sync1
Sync2
control



Polling: checked for next data in/out

INTR: checked by controller to send intr.

8251 yazmaclar

$\overline{CS}$	$C/\overline{D}$	$\overline{RD}$	$\overline{WR}$
1	X	X	X
0	X	1	1
0	1	0	1
0	1	1	0
0	0	0	1
0	0	1	0

Anlam
Data-bus tristate

Data-bus tristate

Status $\rightarrow$ CPU

Mode, Control, Sync $\leftarrow$ CPU

Data $\rightarrow$ CPU

Data $\leftarrow$ CPU

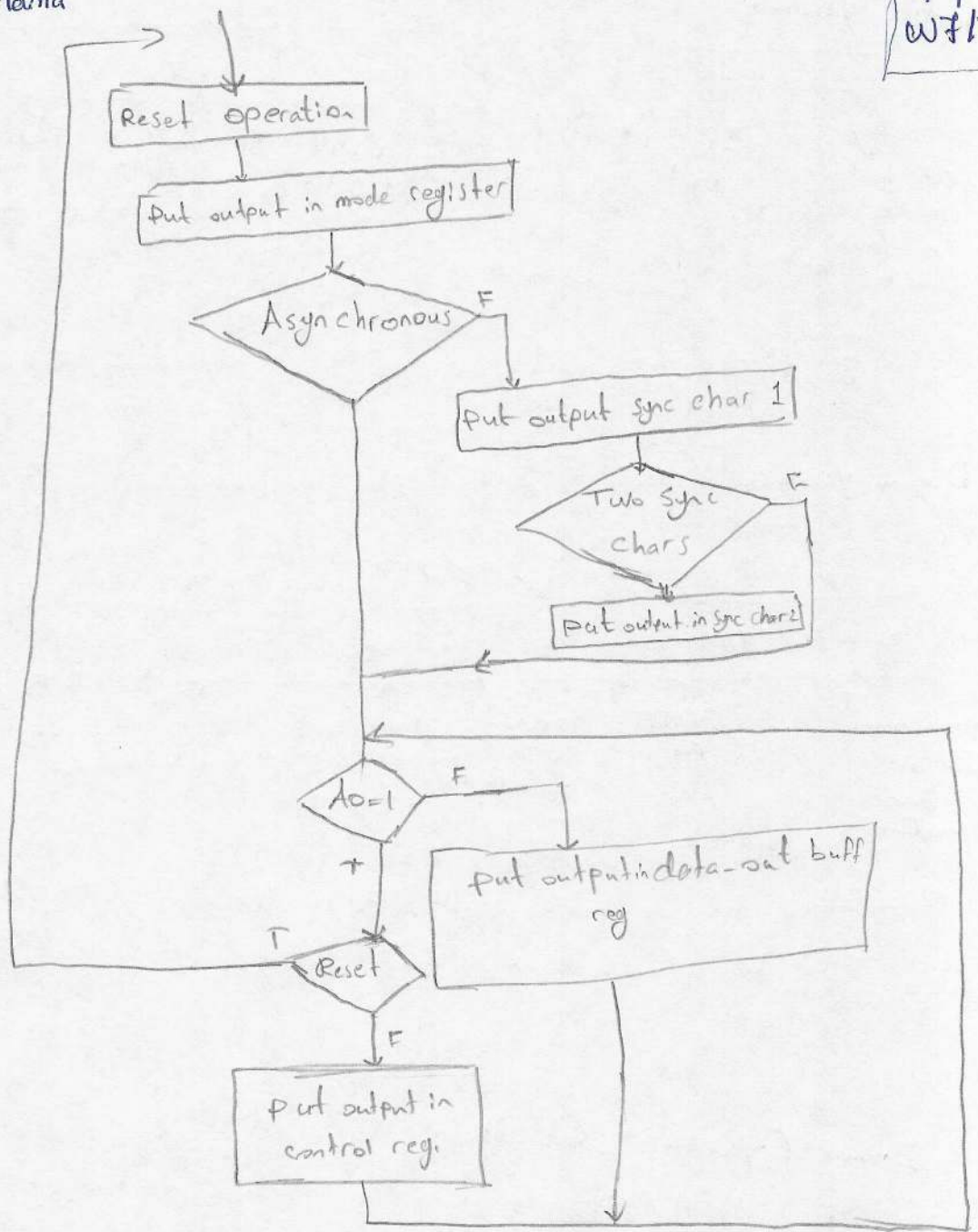
Hattın koparılmış

8251 ilk Ayarlama

mp
w717

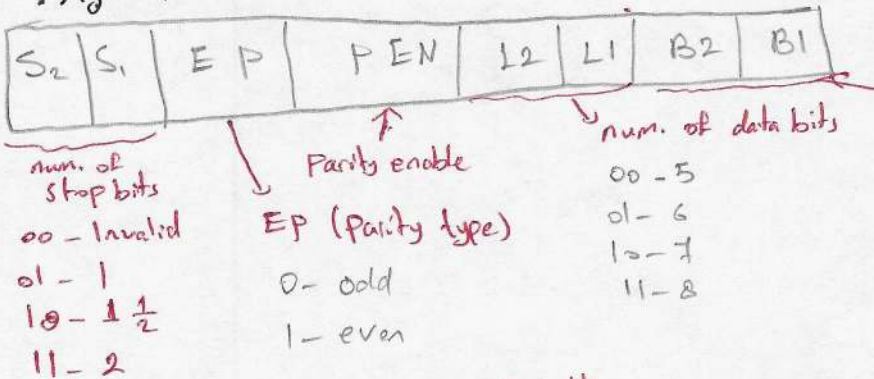
Bizim akışımız:

- Reset
- $C/D = 1, \overline{WR} = \phi \Rightarrow$ mode (asyn)
- $C/D = 1, \overline{WR} = \phi \Rightarrow$ Kontrol (reset) yapılması



Mod yazma:

• Asyn mode

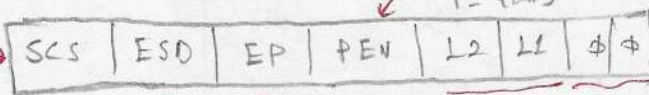


Baud rate Factor
 00 - Sync mode
 01 - Factor = 1
 10 - Factor = 16
 11 - Factor = 64

Elbch
 işaretleri
 b. degerlere
 balseair
 (Pickens olarak)

Sync chars • Sync mod

0 - 2 chars
 1 - 1 char



Ext. Sync. detect: 0 - SYNDET is output, 1 - SYNDET is input
 Parity type: 0 - odd, 1 - Even
 # data bits: 00 - 5, 01 - 6, 10 - 7, 11 - 8

Sync mode
 (when Φ)

Kontrol Yazmacı



Enter hunt mode
(Φ olarak kullanacağız)

Internal reset

all three err bits in status to be cleared

break by forcing TXD low

Seri haberleşme kesiliyor

modemle ilgili

Forces DTR output low

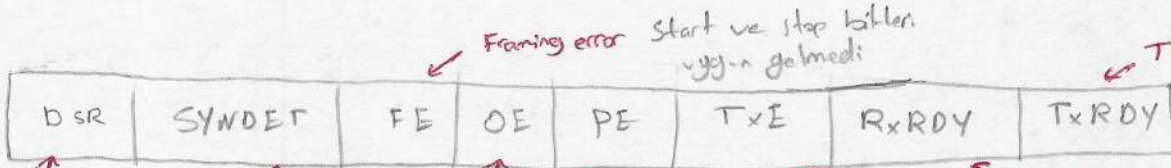
Transmitter enable

Dış dünyaya veri gönderiyorsak



8251 iki farklı şekilde resetlenebilir: 1) Fiziki Reset ucuna 1 göndermek
2) yazılım sal olarak (bkz. *)

Durum Yazmacı (adres ucu 1 iken yaptığımız okumalar sırasında anlamlı)



1 indicates DSR pin is active (modem)

SYNDET pin

overrun error
Veri gönderimini bitlerden veri geldi

Parity error
Gelen Verinin parity biti hatalı

TxE pin
Transmit enable

Transmit ready (i)
Dış dünyaya veri çıkabilir miyim?

RxRDY pin
Dış dünyadan Veri okuyabilir miyim?

(i) indicates data-out buffer register empty; Not affected by CTS and TxE control bit (unlike TxRDY pin)

Example (Yazılım sal Seri iletişimi)

Yazılım sal olarak Parity kullanmadan (9600) baud rate değerinde 8 bit veri aktarımı sağlayan asm. kodunu yaz.

1. Start bit'i gönder $\left(\begin{array}{l} \text{CLC} \\ \text{RCL AL,1} \\ \text{OUT P8,AL} \end{array} \right)$

2. 104 μ s bekle RCR AL,1

9 kere $\left(\begin{array}{l} \text{STC} \\ \text{RCR AL,1} \\ \text{OUT P8,AL} \\ \text{104 } \mu\text{s bekle} \end{array} \right)$